

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 11 · LECTURE NOTES

Fair Division

Dividing without money, and what fairness can mean.

MODULE OUTCOMES

- Define proportionality, envy-freeness, and equitability.
- Explain divide-and-choose and its generalisations.
- Explain the indivisible case and its approximations.
- Explain the complexity of envy-free division.
- Choose a fairness criterion and defend it.

1 The criteria

Criterion	Definition	Note
Proportional	Each of n agents receives at least $1/n$ of the total by its own valuation.	The weakest useful notion, and the easiest to achieve.
Envy-free	No agent prefers another agent's share to its own.	Implies proportionality, and is strictly stronger for three or more agents.
Equitable	Every agent's own valuation of its own share is the same number.	Independent of envy-freeness — neither implies the other, which surprises people.
Efficient (Pareto optimal)	No reallocation makes someone better off without making someone worse off.	Not a fairness notion at all — giving everything to one agent is Pareto optimal. See the note.
Maximin share	At least what you could guarantee by partitioning the items yourself and choosing last.	The natural indivisible analogue of proportionality (§3).

These are genuinely different properties and they can conflict — an allocation can be equitable and envious, or envy-free and inefficient — so 'fair' is not a specification, and naming which one you mean is the first step of any such design. Pareto efficiency is in the table because it is constantly mistaken for a fairness criterion, including in engineering requirements: it constrains waste, not distribution.

2 The divisible case

DIVIDE-AND-CHOOSE IS ENVY-FREE FOR TWO AGENTS AND NEEDS NO KNOWLEDGE OF THE OTHER'S PREFERENCES

One agent cuts and the other chooses — the cutter makes the two pieces equal by its own valuation, and is therefore content with either; the chooser takes whichever it prefers, and is content by construction.

So neither agent envies the other, and crucially neither needed to know anything whatsoever about the other's preferences — which is the dominant-strategy property (Module 02 §2) appearing in a procedural rather than a direct-revelation form, and is why the protocol is three thousand years old and still correct.

And it generalises to n agents for proportionality — moving-knife protocols and the last-diminisher procedure both achieve it — but envy-freeness for n agents is very much harder, and bounded protocols for it were found only recently, which is a striking fact about a problem that sounds elementary.

Which is the shape of the divisible case: proportionality is easy, envy-freeness is hard, and both are achievable — unlike §3's indivisible case, where one of them is simply unavailable.

3 The indivisible case

THE OBSTACLE

one item, two agents, and both want it.
No allocation is envy-free. Not hard -- impossible.

SO THE NOTIONS ARE RELAXED

EF1 envy-free up to one item: any envy disappears if one item is removed from the envied bundle. ALWAYS ACHIEVABLE, by a simple round-robin draft.
EFX envy-free up to ANY item: strictly stronger, and its existence was open for a long time.
MMS maximin share: what you could guarantee yourself by partitioning the items and choosing last. NOT always achievable, which was a genuine surprise; constant factor approximations exist.

AND THE COMPUTATION

maximising welfare subject to a fairness constraint is typically NP-hard, so this is an approximation problem like Module 08's.

PLUS: agents report their valuations, so the incentive problem of Module 06 applies on top.

EF1 is always achievable by a simple round-robin draft — agents take turns picking their favourite remaining item — **which is a rare case of a real fairness guarantee with a trivial algorithm**, and is worth knowing because it is immediately implementable. **Maximin share not always being achievable was a genuine surprise** to the field, since the definition is constructed precisely so that the agent could guarantee it for itself — and the counterexamples are intricate, which is why it took a while.

4 Choosing a criterion

- **Divisible or indivisible?** — which determines whether exact fairness is even available (§3's obstacle), and is the first thing to establish.
- **Which fairness notion, and why that one?** — and are you willing to lose efficiency for it, which you frequently must: the most efficient allocation is rarely the fairest under any of §1's criteria.
- **Is there money?** — side payments make almost everything easier (Module 06 §4, Module 10 §3) — and their absence is usually a policy or legal constraint rather than a technical one, which is worth confirming before designing around it.
- **Are the valuations additive?** — most results in this module assume it, and most real preferences are not (Module 08 §1's complements and substitutes), so check before applying a theorem.

- **And is the mechanism truthful? — most fair division procedures are not, including divide-and-choose in the n -agent generalisations — and that should be stated rather than left for participants to discover. This is rarely stated alongside the fairness guarantees and should be.**

AND 'FAIR' WITHOUT A DEFINITION IS NOT A SPECIFICATION

The criteria in §1 are genuinely different and are sometimes mutually incompatible — so a requirement to 'allocate fairly' has not said what to build, and two engineers will build different things from it.

Which is this module's practical contribution: a vocabulary precise enough to turn a value judgement into a specification that can be checked — and checked automatically, in most cases.

And the choice among the criteria is a value judgement — which the mathematics does not make, and should not be presented as making: a paper that proves its mechanism is EF1 has not shown that EF1 is what anybody wanted.

So state the criterion, state what it costs in efficiency, and state whether the procedure is truthful — three things, and Module 13 wants all of them (as it does for every other module's mechanisms).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Brams & Taylor — Fair Division	§§1 and 2 — the procedures, including the moving-knife constructions. Library copy.
Procaccia — Cake cutting algorithms, in the Handbook (free PDF)	§2 — the computational treatment, including the query complexity of envy-freeness.
Caragiannis et al. — The unreasonable fairness of maximum Nash welfare (free)	§3 — a rule that is simultaneously EF1 and efficient, which is a satisfying result.
Spliddit (free)	§§3 and 4 — these algorithms as a working service, which makes the criteria concrete.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Define all five criteria** and give an allocation satisfying each but not the others.
- 2 **Show that envy-freeness implies proportionality.**

- 3 **Find an equitable allocation with envy**, and the converse.
- 4 **Show that giving everything to one agent is Pareto optimal.**
- 5 **Run divide-and-choose** and verify envy-freeness.
- 6 **Implement a proportional n-agent protocol.**
- 7 **Prove no envy-free allocation exists** for one item and two agents.
- 8 **Implement the round-robin** and verify EF1.
- 9 **Find a maximin share counterexample** in the literature and work through it.
- 10 **Find a profitable misreport** in one fair division procedure.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define five fairness criteria.
- 2 Which implies which, and which are independent?
- 3 Why is Pareto efficiency not a fairness notion?
- 4 Describe divide-and-choose and why it works.
- 5 What does it require knowing about the other agent?
- 6 What generalises to n agents, and what does not?
- 7 Why is envy-freeness impossible with indivisible items?
- 8 Define EF1, EFX, and MMS, and say which is always achievable.
- 9 Give the five design questions.
- 10 Why is 'fair' not a specification?